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Subject: Linear Algebra

Section: TS21

Class Activity 2: Determinants and Inverses

1. Given matrix A

$$A = \begin{bmatrix} 1 & 4 & -2 & 4 & 6 \\ 0 & 3 & -1 & 9 & 5 \\ 3 & 0 & 0 & 0 & -1 \\ 0 & 3 & 4 & 3 & 1 \\ 0 & 4 & 8 & 0 & -3 \end{bmatrix}$$

a. det A using cofactor expansion

Row 3 has three zeros, so expand along row 3: $\det A = a_{31}C_{31} + a_{35}C_{35} = 3C_{31} + (-1)C_{35}$.

$$M_{31} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 4 (it has the most zeros):

$$\det \begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)] = (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\det \begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)] = 4(-6) - 4(-12) + 6(-18) = -84$$

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)] = 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{31} = -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$C_{31} = (-1)^{3+1}M_{31} = 276$$

$$M_{35} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)] = 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{35} = 1(-12) = -12$$

$$C_{35} = (-1)^{3+5}M_{35} = -12$$

$$\det A = 3(276) + (-1)(-12) = 828 + 12 = 840$$

$$\boxed{\det A = 840}$$

b. All the minors M_{ij} of A

M_{ij} is the determinant of the 4×4 matrix left after deleting row i and column j of A. Each one is expanded along its row or column with the most zeros.

$$M_{11} = \begin{vmatrix} 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)] = 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{11} = (-1)(-12) = 12$$

$$M_{12} = \begin{vmatrix} 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} -1 & 9 & 5 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-1)[(3)(-3) - (1)(0)] - 9[(4)(-3) - (1)(8)] + 5[(4)(0) - (3)(8)] = (-1)(-9) - 9(-20) + 5(-24) = 69$$

$$M_{12} = -3(69) = -207$$

$$M_{13} = \begin{vmatrix} 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & 9 & 5 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 3[(3)(-3) - (1)(0)] - 9[(3)(-3) - (1)(4)] + 5[(3)(0) - (3)(4)] = 3(-9) - 9(-13) + 5(-12) = 30$$

$$M_{13} = -3(30) = -90$$

$$M_{14} = \begin{vmatrix} 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 5 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 3[(4)(-3) - (1)(8)] - (-1)[(3)(-3) - (1)(4)] + 5[(3)(8) - (4)(4)] = 3(-20) - (-1)(-13) + 5(8) = -33$$

$$M_{14} = -3(-33) = 99$$

$$M_{15} = \begin{vmatrix} 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)] = 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{15} = -3(-12) = 36$$

$$M_{21} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 4[(4)(0) - (3)(8)] - (-2)[(3)(0) - (3)(4)] + 4[(3)(8) - (4)(4)] = 4(-24) - (-2)(-12) + 4(8) = -88$$

$$M_{21} = (-1)(-88) = 88$$

$$M_{22} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} -2 & 4 & 6 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-2)[(3)(-3) - (1)(0)] - 4[(4)(-3) - (1)(8)] + 6[(4)(0) - (3)(8)] = (-2)(-9) - 4(-20) + 6(-24) = -46$$

$$\det \begin{vmatrix} 1 & -2 & 4 \\ 0 & 4 & 3 \\ 0 & 8 & 0 \end{vmatrix} = 1[(4)(0) - (3)(8)] - (-2)[(0)(0) - (3)(0)] + 4[(0)(8) - (4)(0)] = 1(-24) - (-2)(0) + 4(0) = -24$$

$$M_{22} = -3(-46) + (-1)(-24) = 162$$

$$M_{23} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 4 & 4 & 6 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 4[(3)(-3) - (1)(0)] - 4[(3)(-3) - (1)(4)] + 6[(3)(0) - (3)(4)] = 4(-9) - 4(-13) + 6(-12) = -56$$

$$\det \begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 3 \\ 0 & 4 & 0 \end{vmatrix} = 1[(3)(0) - (3)(4)] - 4[(0)(0) - (3)(0)] + 4[(0)(4) - (3)(0)] = 1(-12) - 4(0) + 4(0) = -12$$

$$M_{23} = -3(-56) + (-1)(-12) = 180$$

$$M_{24} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 6 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 4[(4)(-3) - (1)(8)] - (-2)[(3)(-3) - (1)(4)] + 6[(3)(8) - (4)(4)] = 4(-20) - (-2)(-13) + 6(8) = -58$$

$$\det \begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & 4 \\ 0 & 4 & 8 \end{vmatrix} = 1[(3)(8) - (4)(4)] - 4[(0)(8) - (4)(0)] + (-2)[(0)(4) - (3)(0)] = 1(8) - 4(0) + (-2)(0) = 8$$

$$M_{24} = -3(-58) + (-1)(8) = 166$$

$$M_{25} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix}$$

Expand along row 2 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 4[(4)(0) - (3)(8)] - (-2)[(3)(0) - (3)(4)] + 4[(3)(8) - (4)(4)] = 4(-24) - (-2)(-12) + 4(8) = -88$$

$$M_{25} = -3(-88) = 264$$

$$M_{31} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 3 & 4 & 3 & 1 \\ 4 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 4 (it has the most zeros):

$$\det \begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)] = (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\det \begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)] = 4(-6) - 4(-12) + 6(-18) = -84$$

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)] = 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{31} = -4(-138) + 8(-84) + (-3)(-132) = 276$$

$$M_{32} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 0 & 4 & 3 & 1 \\ 0 & 8 & 0 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} -1 & 9 & 5 \\ 4 & 3 & 1 \\ 8 & 0 & -3 \end{vmatrix} = (-1)[(3)(-3) - (1)(0)] - 9[(4)(-3) - (1)(8)] + 5[(4)(0) - (3)(8)] = (-1)(-9) - 9(-20) + 5(-24) = 69$$

$$M_{32} = 1(69) = 69$$

$$M_{33} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 0 & 3 & 3 & 1 \\ 0 & 4 & 0 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & 9 & 5 \\ 3 & 3 & 1 \\ 4 & 0 & -3 \end{vmatrix} = 3[(3)(-3) - (1)(0)] - 9[(3)(-3) - (1)(4)] + 5[(3)(0) - (3)(4)] = 3(-9) - 9(-13) + 5(-12) = 30$$

$$M_{33} = 1(30) = 30$$

$$M_{34} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 0 & 3 & 4 & 1 \\ 0 & 4 & 8 & -3 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 5 \\ 3 & 4 & 1 \\ 4 & 8 & -3 \end{vmatrix} = 3[(4)(-3) - (1)(8)] - (-1)[(3)(-3) - (1)(4)] + 5[(3)(8) - (4)(4)] = 3(-20) - (-1)(-13) + 5(8) = -33$$

$$M_{34} = 1(-33) = -33$$

$$M_{35} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 0 & 3 & 4 & 3 \\ 0 & 4 & 8 & 0 \end{vmatrix}$$

Expand along column 1 (it has the most zeros):

$$\det \begin{vmatrix} 3 & -1 & 9 \\ 3 & 4 & 3 \\ 4 & 8 & 0 \end{vmatrix} = 3[(4)(0) - (3)(8)] - (-1)[(3)(0) - (3)(4)] + 9[(3)(8) - (4)(4)] = 3(-24) - (-1)(-12) + 9(8) = -12$$

$$M_{35} = 1(-12) = -12$$

$$M_{41} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 4 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 4 & 8 & 0 \end{vmatrix} = 4[(-1)(0) - (9)(8)] - (-2)[(3)(0) - (9)(4)] + 4[(3)(8) - (-1)(4)] = 4(-72) - (-2)(-36) + 4(28) = -248$$

$$M_{41} = -(-1)(-248) = -248$$

$$M_{42} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 8 & 0 & -3 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 8 & 0 & -3 \end{vmatrix} = (-2)[(9)(-3) - (5)(0)] - 4[(-1)(-3) - (5)(8)] + 6[(-1)(0) - (9)(8)] = (-2)(-27) - 4(-37) + 6(-72) = -230$$

$$\det \begin{vmatrix} 1 & -2 & 4 \\ 0 & -1 & 9 \\ 0 & 8 & 0 \end{vmatrix} = 1[(-1)(0) - (9)(8)] - (-2)[(0)(0) - (9)(0)] + 4[(0)(8) - (-1)(0)] = 1(-72) - (-2)(0) + 4(0) = -72$$

$$M_{42} = 3(-230) - (-1)(-72) = -762$$

$$M_{43} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 0 & -3 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 4 & 0 & -3 \end{vmatrix} = 4[(9)(-3) - (5)(0)] - 4[(3)(-3) - (5)(4)] + 6[(3)(0) - (9)(4)] = 4(-27) - 4(-29) + 6(-36) = -208$$

$$\det \begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 9 \\ 0 & 4 & 0 \end{vmatrix} = 1[(3)(0) - (9)(4)] - 4[(0)(0) - (9)(0)] + 4[(0)(4) - (3)(0)] = 1(-36) - 4(0) + 4(0) = -36$$

$$M_{43} = 3(-208) - (-1)(-36) = -660$$

$$M_{44} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 8 & -3 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 6 \\ 3 & -1 & 5 \\ 4 & 8 & -3 \end{vmatrix} = 4[(-1)(-3) - (5)(8)] - (-2)[(3)(-3) - (5)(4)] + 6[(3)(8) - (-1)(4)] = 4(-37) - (-2)(-29) + 6(28) = -38$$

$$\det \begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & -1 \\ 0 & 4 & 8 \end{vmatrix} = 1[(3)(8) - (-1)(4)] - 4[(0)(8) - (-1)(0)] + (-2)[(0)(4) - (3)(0)] = 1(28) - 4(0) + (-2)(0) = 28$$

$$M_{44} = 3(-38) - (-1)(28) = -86$$

$$M_{45} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 4 & 8 & 0 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 4 & 8 & 0 \end{vmatrix} = 4[(-1)(0) - (9)(8)] - (-2)[(3)(0) - (9)(4)] + 4[(3)(8) - (-1)(4)] = 4(-72) - (-2)(-36) + 4(28) = -248$$

$$M_{45} = 3(-248) = -744$$

$$M_{51} = \begin{vmatrix} 4 & -2 & 4 & 6 \\ 3 & -1 & 9 & 5 \\ 0 & 0 & 0 & -1 \\ 3 & 4 & 3 & 1 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)] = 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{51} = -(-1)(-132) = -132$$

$$M_{52} = \begin{vmatrix} 1 & -2 & 4 & 6 \\ 0 & -1 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 4 & 3 & 1 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} -2 & 4 & 6 \\ -1 & 9 & 5 \\ 4 & 3 & 1 \end{vmatrix} = (-2)[(9)(1) - (5)(3)] - 4[(-1)(1) - (5)(4)] + 6[(-1)(3) - (9)(4)] = (-2)(-6) - 4(-21) + 6(-39) = -138$$

$$\det \begin{vmatrix} 1 & -2 & 4 \\ 0 & -1 & 9 \\ 0 & 4 & 3 \end{vmatrix} = 1[(-1)(3) - (9)(4)] - (-2)[(0)(3) - (9)(0)] + 4[(0)(4) - (-1)(0)] = 1(-39) - (-2)(0) + 4(0) = -39$$

$$M_{52} = 3(-138) - (-1)(-39) = -453$$

$$M_{53} = \begin{vmatrix} 1 & 4 & 4 & 6 \\ 0 & 3 & 9 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 3 & 1 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & 4 & 6 \\ 3 & 9 & 5 \\ 3 & 3 & 1 \end{vmatrix} = 4[(9)(1) - (5)(3)] - 4[(3)(1) - (5)(3)] + 6[(3)(3) - (9)(3)] = 4(-6) - 4(-12) + 6(-18) = -84$$

$$\det \begin{vmatrix} 1 & 4 & 4 \\ 0 & 3 & 9 \\ 0 & 3 & 3 \end{vmatrix} = 1[(3)(3) - (9)(3)] - 4[(0)(3) - (9)(0)] + 4[(0)(3) - (3)(0)] = 1(-18) - 4(0) + 4(0) = -18$$

$$M_{53} = 3(-84) - (-1)(-18) = -270$$

$$M_{54} = \begin{vmatrix} 1 & 4 & -2 & 6 \\ 0 & 3 & -1 & 5 \\ 3 & 0 & 0 & -1 \\ 0 & 3 & 4 & 1 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 6 \\ 3 & -1 & 5 \\ 3 & 4 & 1 \end{vmatrix} = 4[(-1)(1) - (5)(4)] - (-2)[(3)(1) - (5)(3)] + 6[(3)(4) - (-1)(3)] = 4(-21) - (-2)(-12) + 6(15) = -18$$

$$\det \begin{vmatrix} 1 & 4 & -2 \\ 0 & 3 & -1 \\ 0 & 3 & 4 \end{vmatrix} = 1[(3)(4) - (-1)(3)] - 4[(0)(4) - (-1)(0)] + (-2)[(0)(3) - (3)(0)] = 1(15) - 4(0) + (-2)(0) = 15$$

$$M_{54} = 3(-18) - (-1)(15) = -39$$

$$M_{55} = \begin{vmatrix} 1 & 4 & -2 & 4 \\ 0 & 3 & -1 & 9 \\ 3 & 0 & 0 & 0 \\ 0 & 3 & 4 & 3 \end{vmatrix}$$

Expand along row 3 (it has the most zeros):

$$\det \begin{vmatrix} 4 & -2 & 4 \\ 3 & -1 & 9 \\ 3 & 4 & 3 \end{vmatrix} = 4[(-1)(3) - (9)(4)] - (-2)[(3)(3) - (9)(3)] + 4[(3)(4) - (-1)(3)] = 4(-39) - (-2)(-18) + 4(15) = -132$$

$$M_{55} = 3(-132) = -396$$

Summary of minors:

$$[M_{ij}] = \begin{bmatrix} 12 & -207 & -90 & 99 & 36 \\ 88 & 162 & 180 & 166 & 264 \\ 276 & 69 & 30 & -33 & -12 \\ -248 & -762 & -660 & -86 & -744 \\ -132 & -453 & -270 & -39 & -396 \end{bmatrix}$$

c. Cofactors C_{ij} of A

$C_{ij} = (-1)^{i+j} M_{ij}$, so the sign follows the checkerboard pattern below and flips the minor wherever $i + j$ is odd.

$$\begin{bmatrix} + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \\ - & + & - & + & - \\ + & - & + & - & + \end{bmatrix}$$

$$\begin{aligned} C_{11} &= + (12) = 12, & C_{12} &= - (-207) = 207, & C_{13} &= + (-90) = -90, & C_{14} &= - (99) = -99, & C_{15} &= + (36) = 36 \\ C_{21} &= - (88) = -88, & C_{22} &= + (162) = 162, & C_{23} &= - (180) = -180, & C_{24} &= + (166) = 166, & C_{25} &= - (264) = -264 \\ C_{31} &= + (276) = 276, & C_{32} &= - (69) = -69, & C_{33} &= + (30) = 30, & C_{34} &= - (-33) = 33, & C_{35} &= + (-12) = -12 \\ C_{41} &= - (-248) = 248, & C_{42} &= + (-762) = -762, & C_{43} &= - (-660) = 660, & C_{44} &= + (-86) = -86, & C_{45} &= - (-744) = 744 \\ C_{51} &= + (-132) = -132, & C_{52} &= - (-453) = 453, & C_{53} &= + (-270) = -270, & C_{54} &= - (-39) = 39, & C_{55} &= + (-396) = -396 \end{aligned}$$

$$[C_{ij}] = \begin{bmatrix} 12 & 207 & -90 & -99 & 36 \\ -88 & 162 & -180 & 166 & -264 \\ 276 & -69 & 30 & 33 & -12 \\ 248 & -762 & 660 & -86 & 744 \\ -132 & 453 & -270 & 39 & -396 \end{bmatrix}$$

d. Adjoint adj A

The adjoint is the transpose of the cofactor matrix: $\text{adj } A = [C_{ij}]^T$.

$$\text{adj } A = \begin{bmatrix} 12 & -88 & 276 & 248 & -132 \\ 207 & 162 & -69 & -762 & 453 \\ -90 & -180 & 30 & 660 & -270 \\ -99 & 166 & 33 & -86 & 39 \\ 36 & -264 & -12 & 744 & -396 \end{bmatrix}$$

e. A^{-1} using Gauss-Jordan elimination

$$[A \mid I] = \left[\begin{array}{ccccc|ccccc} 1 & 4 & -2 & 4 & 6 & 1 & 0 & 0 & 0 & 0 \\ 0 & 3 & -1 & 9 & 5 & 0 & 1 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 & -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 3 & 4 & 3 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & 8 & 0 & -3 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow -3R_1 + R_3 \rightarrow R_3$: clear column 1

$$\left[\begin{array}{ccccc|ccccc} 1 & 4 & -2 & 4 & 6 & 1 & 0 & 0 & 0 & 0 \\ 0 & 3 & -1 & 9 & 5 & 0 & 1 & 0 & 0 & 0 \\ 0 & -12 & 6 & -12 & -19 & -3 & 0 & 1 & 0 & 0 \\ 0 & 3 & 4 & 3 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & 8 & 0 & -3 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow \frac{1}{3}R_2 \rightarrow R_2$: make the pivot 1

$$\left[\begin{array}{ccccc|ccccc} 1 & 4 & -2 & 4 & 6 & 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & 3 & \frac{5}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ 0 & -12 & 6 & -12 & -19 & -3 & 0 & 1 & 0 & 0 \\ 0 & 3 & 4 & 3 & 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 4 & 8 & 0 & -3 & 0 & 0 & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow -4R_2 + R_1 \rightarrow R_1$, $\rightarrow 12R_2 + R_3 \rightarrow R_3$, $\rightarrow -3R_2 + R_4 \rightarrow R_4$, $\rightarrow -4R_2 + R_5 \rightarrow R_5$: clear column 2

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & -\frac{2}{3} & -8 & -\frac{2}{3} & 1 & -\frac{4}{3} & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & 3 & \frac{5}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 2 & 24 & 1 & -3 & 4 & 1 & 0 & 0 \\ 0 & 0 & 5 & -6 & -4 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & \frac{28}{3} & -12 & -\frac{29}{3} & 0 & -\frac{4}{3} & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow \frac{1}{2}R_3 \rightarrow R_3$: make the pivot 1

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & -\frac{2}{3} & -8 & -\frac{2}{3} & 1 & -\frac{4}{3} & 0 & 0 & 0 \\ 0 & 1 & -\frac{1}{3} & 3 & \frac{5}{3} & 0 & \frac{1}{3} & 0 & 0 & 0 \\ 0 & 0 & 1 & 12 & \frac{1}{2} & -\frac{3}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 5 & -6 & -4 & 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & \frac{28}{3} & -12 & -\frac{29}{3} & 0 & -\frac{4}{3} & 0 & 0 & 1 \end{array} \right]$$

$\rightarrow \frac{2}{3}R_3 + R_1 \rightarrow R_1$, $\rightarrow \frac{1}{3}R_3 + R_2 \rightarrow R_2$, $\rightarrow -5R_3 + R_4 \rightarrow R_4$, $\rightarrow -\frac{28}{3}R_3 + R_5 \rightarrow R_5$: clear column 3

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 7 & \frac{11}{6} & -\frac{1}{2} & 1 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 1 & 12 & \frac{1}{2} & -\frac{3}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & -66 & -\frac{13}{2} & \frac{15}{2} & -11 & -\frac{5}{2} & 1 & 0 \\ 0 & 0 & 0 & -124 & -\frac{43}{3} & 14 & -20 & -\frac{14}{3} & 0 & 1 \end{array} \right]$$

$\rightarrow -\frac{1}{66}R_4 \rightarrow R_4$: make the pivot 1

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 7 & \frac{11}{6} & -\frac{1}{2} & 1 & \frac{1}{6} & 0 & 0 \\ 0 & 0 & 1 & 12 & \frac{1}{2} & -\frac{3}{2} & 2 & \frac{1}{2} & 0 & 0 \\ 0 & 0 & 0 & 1 & \frac{13}{132} & -\frac{5}{44} & \frac{1}{6} & \frac{5}{132} & -\frac{1}{66} & 0 \\ 0 & 0 & 0 & -124 & -\frac{43}{3} & 14 & -20 & -\frac{14}{3} & 0 & 1 \end{array} \right]$$

$\rightarrow -7R_4 + R_2 \rightarrow R_2$, $\rightarrow -12R_4 + R_3 \rightarrow R_3$, $\rightarrow 124R_4 + R_5 \rightarrow R_5$: clear column 4

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{151}{132} & \frac{13}{44} & -\frac{1}{6} & -\frac{13}{132} & \frac{7}{66} & 0 \\ 0 & 0 & 1 & 0 & -\frac{15}{22} & -\frac{3}{22} & 0 & \frac{1}{22} & \frac{2}{11} & 0 \\ 0 & 0 & 0 & 1 & \frac{13}{132} & -\frac{5}{44} & \frac{1}{6} & \frac{5}{132} & -\frac{1}{66} & 0 \\ 0 & 0 & 0 & 0 & -\frac{70}{33} & -\frac{1}{11} & \frac{2}{3} & \frac{1}{33} & -\frac{62}{33} & 1 \end{array} \right]$$

$\rightarrow -\frac{33}{70}R_5 \rightarrow R_5$: make the pivot 1

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & -\frac{1}{3} & 0 & 0 & \frac{1}{3} & 0 & 0 \\ 0 & 1 & 0 & 0 & \frac{151}{132} & \frac{13}{44} & -\frac{1}{6} & -\frac{13}{132} & \frac{7}{66} & 0 \\ 0 & 0 & 1 & 0 & -\frac{15}{22} & -\frac{3}{22} & 0 & \frac{1}{22} & \frac{2}{11} & 0 \\ 0 & 0 & 0 & 1 & \frac{13}{132} & -\frac{5}{44} & \frac{1}{6} & \frac{5}{132} & -\frac{1}{66} & 0 \\ 0 & 0 & 0 & 0 & 1 & \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70} \end{array} \right]$$

$\rightarrow \frac{1}{3}R_5 + R_1 \rightarrow R_1$, $\rightarrow -\frac{151}{132}R_5 + R_2 \rightarrow R_2$, $\rightarrow \frac{15}{22}R_5 + R_3 \rightarrow R_3$, $\rightarrow -\frac{13}{132}R_5 + R_4 \rightarrow R_4$: clear column 5

$$\left[\begin{array}{ccccc|ccccc} 1 & 0 & 0 & 0 & 0 & \frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\ 0 & 1 & 0 & 0 & 0 & \frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\ 0 & 0 & 1 & 0 & 0 & -\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\ 0 & 0 & 0 & 1 & 0 & -\frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\ 0 & 0 & 0 & 0 & 1 & \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70} \end{array} \right]$$

$$A^{-1} = \begin{bmatrix} \frac{1}{70} & -\frac{11}{105} & \frac{23}{70} & \frac{31}{105} & -\frac{11}{70} \\ \frac{69}{280} & \frac{27}{140} & -\frac{23}{280} & -\frac{127}{140} & \frac{151}{280} \\ -\frac{3}{28} & -\frac{3}{14} & \frac{1}{28} & \frac{11}{14} & -\frac{9}{28} \\ -\frac{33}{280} & \frac{83}{420} & \frac{11}{280} & -\frac{43}{420} & \frac{13}{280} \\ \frac{3}{70} & -\frac{11}{35} & -\frac{1}{70} & \frac{31}{35} & -\frac{33}{70} \end{bmatrix}$$

The same inverse written with the common denominator 840:

$$A^{-1} = \frac{1}{840} \begin{bmatrix} 12 & -88 & 276 & 248 & -132 \\ 207 & 162 & -69 & -762 & 453 \\ -90 & -180 & 30 & 660 & -270 \\ -99 & 166 & 33 & -86 & 39 \\ 36 & -264 & -12 & 744 & -396 \end{bmatrix}$$

Check: $A^{-1} = \frac{1}{\det A} \operatorname{adj} A = \frac{1}{840} \operatorname{adj} A$ gives the same matrix, and $AA^{-1} = I$.

2. Given matrix B

$$B = \begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & -1 \\ 1 & 3 & -6 \end{bmatrix}$$

a. det B using elementary row operations

Adding a multiple of one row to another row does not change the determinant, so only row replacements are used.

$\rightarrow -2R_1 + R_2 \rightarrow R_2$: eliminate 2 first

$$-2R_1 = (-2, -6, -8) + R_2 = ((2-2) (4-6) (-1-8))$$

$$R_2 = (0, -2, -9)$$

$\rightarrow -R_1 + R_3 \rightarrow R_3$: eliminate 1 next

$$-R_1 = (-1, -3, -4) + R_3 = ((1-1) (3-3) (-6-4))$$

$$R_3 = (0, 0, -10)$$

$$U = \begin{bmatrix} 1 & 3 & 4 \\ 0 & -2 & -9 \\ 0 & 0 & -10 \end{bmatrix} \quad (\text{upper triangular})$$

The determinant of a triangular matrix is the product of its diagonal, and the row replacements kept it equal to det B:

$$\det B = \det U = (1)(-2)(-10) = 20$$

$$\boxed{\det B = 20}$$

b. det B using cofactor expansion

Expand along row 1:

$$\begin{aligned} \det B &= 1 \begin{vmatrix} 4 & -1 \\ 3 & -6 \end{vmatrix} - 3 \begin{vmatrix} 2 & -1 \\ 1 & -6 \end{vmatrix} + 4 \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} \\ &= 1[(4)(-6) - (-1)(3)] - 3[(2)(-6) - (-1)(1)] + 4[(2)(3) - (4)(1)] \\ &= 1(-24 + 3) - 3(-12 + 1) + 4(6 - 4) = -21 + 33 + 8 = 20 \end{aligned}$$

$$\boxed{\det B = 20, \text{ same as in (a)}}$$

c. Adjoint of B

Minors. M_{ij} is the 2×2 determinant left after deleting row i and column j .

$$M_{11} = \begin{vmatrix} 4 & -1 \\ 3 & -6 \end{vmatrix} = (4)(-6) - (-1)(3) = -21, \quad M_{12} = \begin{vmatrix} 2 & -1 \\ 1 & -6 \end{vmatrix} = (2)(-6) - (-1)(1) = -11, \quad M_{13} = \begin{vmatrix} 2 & 4 \\ 1 & 3 \end{vmatrix} = (2)(3) - (4)(1) = 2$$

$$M_{21} = \begin{vmatrix} 3 & 4 \\ 3 & -6 \end{vmatrix} = (3)(-6) - (4)(3) = -30, \quad M_{22} = \begin{vmatrix} 1 & 4 \\ 1 & -6 \end{vmatrix} = (1)(-6) - (4)(1) = -10, \quad M_{23} = \begin{vmatrix} 1 & 3 \\ 1 & 3 \end{vmatrix} = (1)(3) - (3)(1) = 0$$

$$M_{31} = \begin{vmatrix} 3 & 4 \\ 4 & -1 \end{vmatrix} = (3)(-1) - (4)(4) = -19, \quad M_{32} = \begin{vmatrix} 1 & 4 \\ 2 & -1 \end{vmatrix} = (1)(-1) - (4)(2) = -9, \quad M_{33} = \begin{vmatrix} 1 & 3 \\ 2 & 4 \end{vmatrix} = (1)(4) - (3)(2) = -2$$

Cofactors. $C_{ij} = (-1)^{i+j} M_{ij}$:

$$C_{11} = +(-21) = -21, \quad C_{12} = -(-11) = 11, \quad C_{13} = +(2) = 2$$

$$C_{21} = -(-30) = 30, \quad C_{22} = +(-10) = -10, \quad C_{23} = -(0) = 0$$

$$C_{31} = +(-19) = -19, \quad C_{32} = -(-9) = 9, \quad C_{33} = +(-2) = -2$$

Cofactor matrix.

$$[C_{ij}] = \begin{bmatrix} -21 & 11 & 2 \\ 30 & -10 & 0 \\ -19 & 9 & -2 \end{bmatrix}$$

Transpose of the cofactor matrix (rows become columns):

$$\boxed{\text{adj } B = [C_{ij}]^T = \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix}}$$

d. Inverse of B using Gauss-Jordan elimination

$$[B \mid I] = \left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 2 & 4 & -1 & 0 & 1 & 0 \\ 1 & 3 & -6 & 0 & 0 & 1 \end{array} \right]$$

$$\rightarrow -2R_1 + R_2 \rightarrow R_2$$

$$-2R_1 = (-2, -6, -8, -2, 0, 0) + R_2 = ((2-2)(4-6)(-1-8)(0-2)(1+0)(0+0))$$

$$R_2 = (0, -2, -9, -2, 1, 0)$$

$$\rightarrow -R_1 + R_3 \rightarrow R_3$$

$$-R_1 = (-1, -3, -4, -1, 0, 0) + R_3 = ((1-1)(3-3)(-6-4)(0-1)(0+0)(1+0))$$

$$R_3 = (0, 0, -10, -1, 0, 1)$$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 0 & -2 & -9 & -2 & 1 & 0 \\ 0 & 0 & -10 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow -\frac{1}{2}R_2 \rightarrow R_2 : \text{make the pivot 1}$$

$$R_2 = (0, 1, \frac{9}{2}, 1, -\frac{1}{2}, 0)$$

$$\left[\begin{array}{ccc|ccc} 1 & 3 & 4 & 1 & 0 & 0 \\ 0 & 1 & \frac{9}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -10 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow -3R_2 + R_1 \rightarrow R_1$$

$$-3R_2 = (0, -3, -\frac{27}{2}, -3, \frac{3}{2}, 0) + R_1 = ((1+0)(3-3)(4-\frac{27}{2})(1-3)(0+\frac{3}{2})(0+0))$$

$$R_1 = (1, 0, -\frac{19}{2}, -2, \frac{3}{2}, 0)$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{19}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & \frac{9}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & -10 & -1 & 0 & 1 \end{array} \right]$$

$$\rightarrow -\frac{1}{10}R_3 \rightarrow R_3 : \text{make the pivot 1}$$

$$R_3 = (0, 0, 1, \frac{1}{10}, 0, -\frac{1}{10})$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -\frac{19}{2} & -2 & \frac{3}{2} & 0 \\ 0 & 1 & \frac{9}{2} & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & \frac{1}{10} & 0 & -\frac{1}{10} \end{array} \right]$$

$$\rightarrow \frac{19}{2}R_3 + R_1 \rightarrow R_1$$

$$\frac{19}{2}R_3 = (0, 0, \frac{19}{2}, \frac{19}{20}, 0, -\frac{19}{20}) + R_1 = ((1+0)(0+0)(-\frac{19}{2}+\frac{19}{2})(-2+\frac{19}{20})(\frac{3}{2}+0)(0-\frac{19}{20}))$$

$$R_1 = (1, 0, 0, -\frac{21}{20}, \frac{3}{2}, -\frac{19}{20})$$

$$\rightarrow -\frac{9}{2}R_3 + R_2 \rightarrow R_2$$

$$-\frac{9}{2}R_3 = (0, 0, -\frac{9}{2}, -\frac{9}{20}, 0, \frac{9}{20}) + R_2 = ((0+0)(1+0)(\frac{9}{2}-\frac{9}{2})(1-\frac{9}{20})(-\frac{1}{2}+0)(0+\frac{9}{20}))$$

$$R_2 = (0, 1, 0, \frac{11}{20}, -\frac{1}{2}, \frac{9}{20})$$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ 0 & 1 & 0 & \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ 0 & 0 & 1 & \frac{1}{10} & 0 & -\frac{1}{10} \end{array} \right]$$

$$B^{-1} = \left[\begin{array}{ccc} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{array} \right]$$

Verification using the adjoint and the determinant:

$$B^{-1} = \frac{1}{\det B} \operatorname{adj} B = \frac{1}{20} \begin{bmatrix} -21 & 30 & -19 \\ 11 & -10 & 9 \\ 2 & 0 & -2 \end{bmatrix} = \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix}$$

Both methods give the same inverse, and $BB^{-1} = I$:

$$\begin{bmatrix} 1 & 3 & 4 \\ 2 & 4 & -1 \\ 1 & 3 & -6 \end{bmatrix} \begin{bmatrix} -\frac{21}{20} & \frac{3}{2} & -\frac{19}{20} \\ \frac{11}{20} & -\frac{1}{2} & \frac{9}{20} \\ \frac{1}{10} & 0 & -\frac{1}{10} \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$